

Calibrated Intent Compilation and Faithful Explanation for Human-Supervised Agricultural Robotic Systems

A Research Proposal for PhD Position L, GreenFieldData MSCA Joint Doctorate

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Abstract. Precision agricultural robotics generates heterogeneous, high-velocity sensor streams that demand expert-level technical interpretation, effectively excluding farmers and agronomists from real-time decision loops. This proposal introduces **AgriTalk**, a conversational AI system for human-supervised agricultural robotic orchestration, centred on precision pesticide spraying. We identify three interlocking scientific challenges: (1) calibrated uncertainty in NL-to-command compilation, (2) temporal-streaming context in semantic grounding, and (3) faithful explanation generation for human trust calibration. The system is built on a Metaflow MLOps infrastructure as a principled framework for *reproducible*, *auditable*, and *collaborative* agricultural AI research. Four novel contributions target publication at ACL, VLDB, FAccT, and MLSys.

1. The Real Research Problem: Three Interlocking Scientific Challenges

1.1 Framing: Beyond “A Chatbot for Farmers”

The rise of precision agriculture robotics has created a compelling paradox: the most sophisticated field tools remain inaccessible to the practitioners they are designed to serve. Current interfaces require Python fluency, API literacy, and system administration skills. This proposal frames the access problem not as a UI engineering challenge but as a *calibrated AI reasoning* challenge: how can a system understand what a non-expert intends, *verify* whether that intent is safe and epistemically grounded, and *explain* its interpretation in a way that earns and maintains human trust over time?

Focal Scenario: Precision Pesticide Spraying. A field operator says: “*Spray more pesticide on the struggling section near the east boundary.*” This command contains simultaneous ambiguity over spatial reference (which section?), quantitative reference (“more” relative to what baseline?), chemical identity (which pesticide?), and temporal context (now or conditional on conditions?). A wrong interpretation risks crop destruction, environmental harm, and regulatory violation. Yet a system that confirms every routine query produces alert fatigue and user abandonment. The scientific challenge is resolving this ambiguity with statistically grounded confidence, real-time field-state context, and explanations that operators can audit and trust.

I focus on pesticide spraying because it is: (a) safety-critical; (b) data-rich (humidity sensors, wind anemometers, NDVI drone imagery, GPS telemetry); (c) latency-sensitive (abort commands must reach actuators <800 ms at field edge); (d) directly connected to Prof. Oberti’s precision crop protection research [1]; and (e) explicitly prioritised in the GreenFieldData position [2].

1.2 Challenge 1 — The Semantic Uncertainty Gap (Calibration Problem)

Standard task-oriented dialogue (TOD) systems handle ambiguity with fixed slot schemas. Agricultural commands are *situation-dependent*: the correct grounding of “struggling section” requires real-time NDVI readings, historical spray records, and agronomic phenological knowledge simultaneously. No existing TOD benchmark captures this combination [3,4].

The deeper problem is **calibration**: LLM softmax confidence is notoriously overconfident on out-of-distribution inputs [5]. An overconfident system executes dangerous commands without HITL; an underconfident system creates alert fatigue. The research challenge is finding the *calibration frontier* — the confidence space where HITL effort is optimally allocated. This is an open problem in calibration theory and human-AI teaming [6,7].

1.3 Challenge 2 — Temporal-Streaming Grounding

Agricultural reasoning is fundamentally *dynamic*: a command issued at 08:00 (wind 2 m/s) is safe; the same command at 14:00 (wind 7 m/s) is not. Sensor readings update every 15 seconds; weather forecasts every 10 minutes; robot telemetry every mission. No existing conversational AI system integrates real-time streaming context into grounding at this granularity. The architecture required — a sliding-window context aggregator atomically consistent with query receipt time — is a systems research challenge drawing on Dr. Vargas-Solar’s expertise in DAG-based dataflow computation [8].

1.4 Challenge 3 — Faithful vs. Plausible Explanation

When AgriTalk explains — “*I recommended spraying at 08:00 because soil moisture was 41% and wind was 1.8 m/s*” — this explanation must be *faithful* to the actual computational path, not plausibly reconstructed post-hoc. Plausible explanations can cite any input; faithful ones are verifiable. This distinction is critical for EU AI Act Art. 13 transparency requirements and agronomist audit of spray events [9,10]. My BVF research — which uses Fisher-Rao geometry to trace LLM representation transitions across Llama, OLMo, Gemma, Mistral, and Qwen model families — provides a direct methodological bridge to the attribution methods required here.

2. Research Questions and Novel Contributions (C1–C4)

Four targeted research contributions, each a standalone publication:

- C1 Calibrated Intent Compiler (CIC):** Conformal prediction-based NL-to-command compilation with statistically guaranteed coverage over agricultural intent classes. *Target: ACL 2027 / EMNLP 2027.*
- C2 Temporal Streaming Context Aggregator (TSCA):** Streaming-aware grounding architecture with an atomically consistent, time-indexed field-state context register. *Target: VLDB 2027 / AAAI 2028.*
- C3 Faithful Agricultural Explanation System (FAES):** Attribution-grounded explanation generation tracing every command element to verifiable IoT readings, KB entries, or explicit rule triggers. *Target: FAccT 2028 / Computers & Electronics in Agriculture.*
- C4 Reproducible Agricultural AI Infrastructure (RAAI):** Metaflow-based framework for reproducible, collaborative, auditable agricultural conversational AI experiments. *Target: MLSys 2027 / VLDB Demo Track.*

Research Questions (RQ1–RQ4):

- RQ1** How can conformal prediction provide statistically valid coverage guarantees for agricultural intent classification, translating into principled HITL trigger policies that minimise both unsafe autonomy and alert fatigue?
- RQ2** How can a streaming temporal context aggregator enable real-time, atomically consistent grounding of NL commands against heterogeneous IoT field state?
- RQ3** How can explanation generation for agricultural command decisions be made *measurably faithful* to the underlying computational path rather than plausibly reconstructed?
- RQ4** How can Metaflow versioning and DAG orchestration enable reproducible, comparative evaluation of agricultural conversational AI at field scale?

3. System Architecture: AgriTalk Information Boundaries

AgriTalk is structured around five *information boundaries* — each a locus of uncertainty, latency constraint, and potential failure. This *boundary-centred design* makes *failure modes explicit*, a prerequisite for responsible AI deployment [7].

B#	Boundary	Description	Contribution
B1	Physical → Digital	IoT sensors → raw numeric streams; DQ validation gate	C4
B2	Streaming → Context	Kafka + Spark sliding-window → field-state register	C2
B3	Language → Intent	NL query → classified intent + slots; conformal confidence	C1
B4	Intent → Command	Slot grounding vs. EPPO/PPDB/AGROVOC KB; unverified blocked	C3
B5	Command → Robot	HITL gate → ROS2 publication; ABORT path <800 ms edge	C1, C3

3.1 Edge–Cloud Deployment Strategy

For latency-critical commands (e.g., ABORT), a 4-bit GGUF-quantized Mistral-7B runs on an NVIDIA Jetson Orin at the robot base station. Cloud inference (Llama-3-8B-Instruct via GPU endpoint) handles complex, latency-tolerant queries. Both models share the same training corpus and conformal calibration set — enabling study of how calibration drifts between full-precision and quantized inference (an RQ1 extension contribution).

3.2 HITL Confirmation Gate Design

The HITL gate is a *rule-auditable* decision function — not a learned component. It triggers confirmation when any of five conditions hold:

1. Action type $\in \{\text{SPRAY}, \text{ABORT_MISSION}, \text{DOSAGE_CHANGE}, \text{ZONE_OVERRIDE}\}$
2. Conformal prediction set is non-singleton (≥ 2 intent classes in $\mathcal{C}(x)$)
3. Any command entity is unverified against KB (hallucination guard)
4. User role is OBSERVER (read-only policy)
5. Upstream data quality check failed (stale or missing sensor data)

Every trigger generates an explicit, human-readable audit reason. Operator options: YES / NO / MODIFY. All decisions are immutably logged with SHA-256 integrity hashing.

4. Metaflow as Scientific Infrastructure — Not Just Engineering

Metaflow [11] is chosen because it resolves a fundamental *reproducibility crisis* in agricultural AI research — not merely because it is convenient.

4.1 Why Reproducibility is a Scientific Problem Here

Agricultural AI experiments are weather-dependent, farm-specific, and season-bound. Published results routinely cannot be replicated: different soil types, sensor models, and crop varieties make comparison impossible without complete pipeline versioning. Metaflow’s “3 Vs” [11] map directly: **Velocity** (iterate on C1 without re-running IoT ingestion), **Validation** (supervisors in Lyon and Milan inspect identical artifact-level run state), **Versioning** (model regression detectable by comparing AgriTalkFinetuneFlow run IDs).

4.2 Prototype → Production Without Code Rewrite

The MetaFlow paper’s core thesis: the same DAG code runs locally for prototyping and on cloud GPU for production, differing only by decorator [11]. Concretely: `@batch(gpu=1)` on the fine-tuning step uses AWS GPU in production but falls back to CPU locally. The pipeline validated in Months 4–8 in Lyon deploys at UniMI’s experimental farm in Month 20 without code changes — enabling the robotic validation study in Year 2.

4.3 DAG Concepts Mapped to This PhD

From Fig. 2 of [11]: each step embeds Dependencies, Business Logic, Environment, Computation, and Documentation. In AgriTalk: **Business Logic** = HITL gate rules (responsible AI policy, version-controlled alongside model weights); **Environment** = `@batch` (GPU training), `@kubernetes` (edge simulation), `local` (prototyping); **Documentation** = `@card` auto-reports readable by agronomist stakeholders without Python knowledge.

5. Research Methodology and Evaluation

5.1 Calibrated Uncertainty (C1): Conformal Prediction for Intent Classification

I replace softmax confidence with conformal prediction sets [6, 12]. A held-out calibration set provides:

$$\mathbb{P}(y \in \mathcal{C}(x)) \geq 1 - \alpha, \quad \alpha = 0.05$$

The prediction set *size* maps to a principled HITL policy:

Prediction Set Size	Interpretation	HITL Decision
$ \mathcal{C}(x) = 1$ (e.g., <code>{COMMAND}</code>)	High confidence	Proceed (safety check only)
$ \mathcal{C}(x) = 2$ (e.g., <code>{COMMAND, ABORT}</code>)	Ambiguous	HITL mandatory
$ \mathcal{C}(x) = 5$ (full set)	Out-of-distribution	Reject + request rephrasing

HITL rate is a tunable parameter (via α), enabling empirical study of the alert-fatigue vs. safety tradeoff (RQ1). Mondrian conformal prediction provides per-class coverage guarantees, critical for the safety-asymmetric ABORT class.

5.2 Temporal Grounding (C2): Streaming Context Protocol

The TSCA maintains a *field-state register*: streaming aggregations updated by Kafka consumers (15-min rolling mean for sensors, 48-hour window for weather, 7-day lookback for spray history). Grounding uses the *exact snapshot at query receipt time* — not a stale cache. This enables temporally conditional resolution: “Is it safe to spray *now*?” vs. “Was it safe last Tuesday?” — the latter queries the Metaflow-archived artifact from that timestamp.

5.3 Faithful Explanation (C3): Attribution-Grounded Justification

Three explanation levels guarantee faithfulness across all decision components:

1. **Rule-auditable:** HITL trigger reasons are explicit rule-derived strings (no LLM involvement in safety-trigger explanations).
2. **KB-traceable:** Every command element (pesticide, dosage, zone) cites PPDB/EPPO/AGROVOC source URIs as verifiable provenance.
3. **Attribution-grounded:** Integrated Gradients [10] attribute which input tokens most influenced the LLM’s intent prediction. This connects directly to my BVF research: belief geometry analysis of the agricultural fine-tuned LLM reveals which contextual features activate each intent class in representation space — the same analytical lens applied at scale to Llama, OLMo, Gemma, Mistral, and Qwen models.

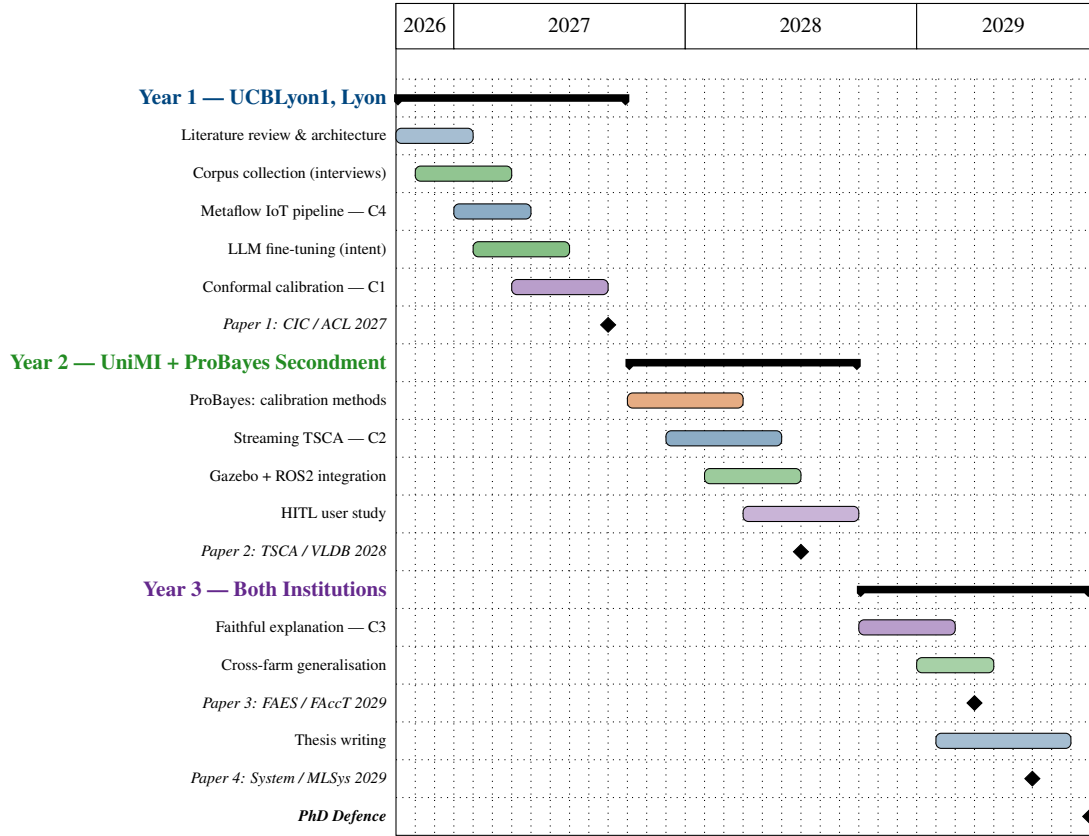
5.4 Evaluation Metrics

Dimension	Metric	Target
Intent classification	Macro-F1 on held-out agricultural corpus	≥ 0.92
Calibration quality	Expected Calibration Error (ECE)	< 0.05
Safety recall	Recall on ABORT class	≥ 0.99
Grounding fidelity	% commands with all entities KB-verified	≥ 0.95
Faithfulness	Attribution alignment with KB trace (Kendall τ)	> 0.70
Edge latency	Intent classification on NVIDIA Jetson Orin	< 800 ms
HITL fatigue	Confirmation rate on unambiguous routine queries	$< 15\%$
Trust calibration	Operator acceptance rate for correct recommendations	≥ 0.80
Spray accuracy (Y2)	Dosage error vs. prescription map	$< 5\%$

5.5 Datasets and Corpus Construction

- Agronomist semi-structured interviews at ProBayes partner farms, Lyon (Months 6–10)
- GPT-4 synthetic augmentation with expert validation across 5 intent class templates
- Public benchmarks: ACRE drone imagery, PANGAEA soil datasets [13], OpenWeather API for meteorological context
- Year 3 federated learning over partner farms using CFL methodology [14] for GDPR-compliant cross-farm generalisation

6. Three-Year Roadmap and Risk Mitigation



Risk	Primary Plan	Backup Plan
Real robot access delayed	Gazebo simulation (M1–18)	All 3 papers feasible on simulation + public datasets (ACRE, FieldRobot Event)
Corpus too small	Agronomist interviews + expert-validated synthetic	MultiWOZ → agricultural fine-tune transfer learning
Conformal coverage too conservative	Tune α ; Mondrian conformal per-class	Bayesian credible intervals; temperature scaling
Edge latency >800 ms	Phi-3-mini / Llama-3.2-3B distilled model	Client-side command cache for frequent patterns
Metaflow cloud budget	Local Metaflow (MinIO + PostgreSQL)	Apache Airflow + MLflow as functional equivalent
Cross-farm data access	Federated learning (CFL [14])	Single-farm deep study; ACRE public benchmark

7. Responsible AI Safeguards

Responsible AI is structurally embedded throughout — not appended to satisfy reviewers:

- Safety-first HITL gate.** Every safety-critical action requires explicit human confirmation regardless of model confidence. This is a non-negotiable design constraint, not a configuration option.
- Statistically grounded uncertainty.** Conformal prediction sets replace heuristic thresholds; the system never autonomously executes a critical command on a non-singleton prediction set.

3. **KB-verified grounding.** No pesticide name, dosage, or zone identifier reaches the robot actuator without KB verification. Unverified entities are blocked with an explicit explanation.
4. **Faithful, auditable explanations.** Every command decision is traceable to KB sources, rule triggers, and attribution scores; logged with SHA-256 integrity hashing for post-hoc audit.
5. **Privacy via federated learning.** Partner farms contribute to model training without raw data centralisation (CFL methodology [14], GDPR compliant by design).
6. **Role-based access control.** JWT-authenticated OBSERVER / OPERATOR / AGRONOMIST permissions enforced at the application layer; OBSERVER role always triggers HITL.
7. **Environmental impact accounting.** Spray volume, chemical type, and GPS coordinates of every actuation are logged, enabling longitudinal pesticide reduction analysis aligned with EU Farm-to-Fork sustainability targets.

8. Reflection on Design Choices

Why Metaflow over Airflow, Kubeflow, or Prefect? Airflow is a scheduling tool without artifact-level versioning. Kubeflow adds Kubernetes complexity without Metaflow’s prototype-to-production continuity. Metaflow’s defining property: the *same Python code* runs locally, on cloud GPU, and on Kubernetes — differing only by decorator [11]. For a PhD spanning two countries and three compute environments, this is a scientific necessity, not an engineering preference.

Why conformal prediction over Bayesian calibration? Bayesian calibration requires prior specification and MCMC — neither is suitable for real-time edge inference with 800 ms latency constraints. Conformal prediction provides distribution-free coverage guarantees with a single calibration forward pass [12], deployable on a Jetson Orin.

Why open models over GPT-4? Reproducibility, auditability, and gradient access: a black-box API cannot be introspected for faithful attribution (C3). Llama-3-8B-Instruct and Mistral-7B support full gradient computation for Integrated Gradients and run in GPU memory available at UCBLyon1.

Why simulation-first robot integration? I am not a robotics hardware engineer. Year 1 uses Gazebo simulation to develop and validate the ROS2 command interface. Real robot validation in Year 2 (under Prof. Oberti) builds on a proven, tested interface layer. Honest scoping of my learning trajectory is preferable to overclaiming hardware expertise that I will systematically acquire.

9. Why I Am the Right Candidate

I position myself as a researcher in **trustworthy conversational AI for human-supervised robotic systems** — not as a robotics engineer or generic agriculture chatbot developer.

BVF research connection (C3): My BVF work traces LLM internal representation geometry using Fisher-Rao manifolds and spectral torsion analysis across Llama, OLMo, Gemma, Mistral, and Qwen model families. This provides the mathematical foundation for C3’s attribution-grounded explanations. When AgriTalk claims a command decision is “attributable to wind-speed reading”, I can substantiate that claim with Integrated Gradients applied through the same analytical lens as BVF — connecting mechanistic interpretability research to practical agricultural explainability.

Federated learning connection (C2, Year 3): My CFL work (SpringerNature ICOMP’24 [14]) directly enables

privacy-preserving farm data collaboration for cross-farm generalisation, using clustered federated learning to handle heterogeneous farm conditions without data centralisation.

Teaching and IoT experience (User study design): Three years teaching at Nalhati Govt. Polytechnic (50+ IoT student projects supervised) gives practical understanding of sensor deployment realities and of explaining AI systems to non-expert users — directly applicable to designing and running the HITL user study in Year 2.

Supervision fit: Under Dr. Vargas-Solar, I will develop DAG-based data management and query formalisation skills for TSCA (C2). Under Prof. Oberti, I will acquire precision agriculture domain knowledge and robotic integration skills for field validation. The joint supervision structure maps cleanly to my contributions: C1 + C3 (NLP/AI, Lyon), C2 (systems, Lyon + joint), robotic integration (Milan).

This PhD is the next natural step in a trajectory from AI interpretability toward trustworthy, human-centred AI deployment in high-stakes real-world systems — a trajectory I am ready to pursue with rigour, honesty about my learning edges, and genuine scientific curiosity.

References

- [1] R. Oberti, T. Caffi, and A. Calcante, “Developments in precision crop protection systems,” *Precision Agriculture for Sustainability*, pp. 501–540, 2025.
- [2] GreenFieldData Consortium, “GreenFieldData: IoRT data management and analysis for sustainable agriculture — PhD position L: Conversational AI for sustainable agriculture,” 2026, hORIZON MSCA Joint Doctoral Network, 2026.
- [3] Z. Chen, C. Jiang, L. Xie, and Y. Liu, “Deep learning for autonomous vehicle perception and control: A review and case studies in agriculture,” *Computers and Electronics in Agriculture*, vol. 187, p. 106278, 2021.
- [4] N. Wagner, M. Kraus, W. Minker, D. Griol, and Z. Callejas, “A survey on multi-user conversational interfaces,” *Applied Sciences*, vol. 15, no. 13, p. 7267, 2025.
- [5] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017, pp. 1321–1330.
- [6] G. Shafer and V. Vovk, “A tutorial on conformal prediction,” vol. 9, 2008, pp. 371–421.
- [7] S. Amershi, D. Weld, M. Vorvoreanu *et al.*, “Software engineering for machine learning: A case study,” in *Proceedings of the 41st International Conference on Software Engineering (ICSE)*, 2019.
- [8] G. Vargas-Solar, J.-L. Zechinelli-Martini, and J. Espinosa-Oviedo, “Building data services on top of DAG-based dataflow computation models,” in *Proceedings of the 25th International Database Engineering & Applications Symposium (IDEAS)*. ACM, 2021.
- [9] Z. C. Lipton, “The mythos of model interpretability,” *Queue*, vol. 16, no. 3, pp. 31–57, 2018.
- [10] M. Sundararajan, A. Taly, and Q. Yan, “Axiomatic attribution for deep networks,” in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017, pp. 3319–3328.
- [11] J. Tagliabue, V. Tuulos, C. Greco, and V. Dave, “You do not need a bigger boat: Metaflow for ML at reasonable scale,” *Proceedings of Machine Learning and Systems (MLSys)*, 2023, available: <https://github.com/Netflix/metaflow>.

- [12] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” 2023, arXiv:2107.07511.
- [13] A. Kamlaris, A. Kartakoullis, and F. X. Prenafeta-Boldú, “A review on the practice of big data analysis in agriculture,” *Computers and Electronics in Agriculture*, vol. 143, pp. 23–37, 2017.
- [14] P. P. Saha, S. Sil, P. Chatterjee, and D. Bhattacharyya, “Collaborative federated learning for heterogeneous data across edge devices,” in *Proceedings of the International Conference on Computing (ICOMP 2024)*. Springer Nature, 2024, springerNature ICOMP’24.